

Unit

1

Lesson

3

Spark The Challenge!

Designing an Inclusive Playground.

Time Frame: [90 min]



Lesson Objectives:

By the end of this lesson students will be able to:

- Identify and use 2D shapes and patterns to design parts of a playground.
- Use bright colors and clear shapes to help all kids, including those who can't see well, play safely.
- Place and change the size of shapes like cubes, balls, and tubes.
- Draw and sort ideas using pictures, simple charts, or diagrams.
- Move around and click shapes in Tinkercad.
- Build playground parts on Tinkercad using shapes and colors that include everyone.
- Understand that different units of measurement (centimeters and meters) are used to measure different sizes of objects.

Challenge questions of the lesson

1. Imagine someone visiting your playground for the very first time. How would your design help them understand where to go and what they can do?
2. What is the most important thing to remember when designing a playground for all our friends

SEL Relation

- Communication (Sharing our ideas): We will practice sharing our designs and explaining our thinking to others.
- Respect (Listening to others): We will listen carefully and respectfully when our classmates present their work.
- Empathy (Thinking about others' feelings): We will think about how our playground designs can make everyone feel included and happy.
- Self-Awareness (Understanding our own work): We will reflect on the choices we made in our design and why they are important.

Engage



Teacher Role:

- **Story Connector:**

Begin the session by reading the short dialogue story aloud to re-engage students with the idea of collaborative playground design.

- **Activator of Prior Knowledge:**

Ask guiding questions to connect to past experiences:

- “Do you remember using Tinkercad or shapes to build models before?”

“What did we learn about ramps, signs, and safe play?”

- **Facilitator:**

Explain the shape activity clearly and distribute the shape cut-out sheets. Guide students to think creatively about how shapes can represent real playground features.

- **Encourager of Design Thinking:**

Prompt students to think like young designers:

- “What can these shapes become? Could this be a ramp, a seat, or a swing part?”

- **Supporter of Expression:**

Allow multiple ways for students to express their ideas—through cutting and pasting, coloring, drawing additions, and explaining their designs to peers.

- Guide students to use a simple AI app such as AI Doodles from Sketch to Art or ChatGPT with teacher assistance.

- Emphasize that AI is a creative partner — it helps visualize their ideas but does not replace their own imagination.

- Encourage safe and guided use (teacher-led demo or small group activity).

Expected From Students:

- **Active Listening:**

Listen to the story and imagine being part of the design team with Miss Sarah, Adam, and Laila.

- **Creative Thinking:**

Think about what different shapes remind them of and how they could be part of a playground design.

- **Hands-On Making:**

Cut out the shapes carefully and create a new playground element, structure, or sign on a separate sheet using the shapes.

- **Application of Learning:**

Use prior knowledge of inclusive design (e.g., ramps, wide spaces, sensory paths) to make their creation helpful and fun for all kids.

- **Visual Communication:**

Color and organize their design clearly so others can understand what they made and how it helps everyone play.

- **AI Exploration (Extension):**

Bring their sketch to life by using an AI tool (with teacher guidance), compare the AI-generated version with their own design, and reflect on what new ideas or improvements it suggests.

- **Reflection & Sharing:**

Talk about their design and the AI version with classmates, explaining how their idea makes the playground more fun, safe, and inclusive.

Parts of the books included.



Story



1. Set the Scene with Excitement and Purpose

- Teacher says (enthusiastically):

"Miss Sarah's class is back today—and they're ready to build something amazing! Can you guess what it is?"

- Ask students:

"If you could build a perfect playground, what would you put in it?"

- (Let students respond.)

This helps activate their imagination and connect personally to the topic.

2. Read the Story Aloud with Voice and Energy

Use tone, facial expressions, and character voices to bring the story to life:

- Use a thoughtful voice for Miss Sarah
- Use an excited, curious voice for Adam and Laila
- Use a wondering tone for Omar

3. Ask a Think-Aloud Question to Prompt Engagement

- Teacher says:

“Wow! Laila and Adam are thinking like real engineers. What do you think they should add to make sure everyone can play?”

- Encourage student responses like “a ramp,” “soft ground,” “a quiet space,” etc.

4. Transition to the Shape Activity

- Teacher says:

“You’re going to be like Miss Sarah’s class today! Let’s start designing using shapes. You’ll get to create something that could go in a real playground!”

- Show the paper with shapes and explain the activity steps briefly before passing them out.

Possible Strategies for Implementation:

1. Character Voice Acting

- Use different voices for each character in the story (Miss Sarah, Adam, Laila, Omar) to bring the dialogue to life.
- Encourage students to echo or repeat lines with expression.

2. Use Puppets or Props

- Create simple stick puppets or paper cut-outs for each character.
- As you read, hold up the correct puppet to help students follow who is speaking.

3. Story Retelling with Student Volunteers

- After reading the story, invite 3–4 students to act out the dialogue.
- Assign roles and let them “perform” the scene with simple gestures or props.

4. Interactive Pause and Predict

- Pause during key lines in the story and ask:
“What do you think Adam will say next?”
“Why do you think Miss Sarah asked about a design program?”

Let's Think!



Objective: Students will explore how geometric shapes can be used to design creative and inclusive playground features. They will cut, arrange, and glue shapes to build their own design, then extend their creativity by using AI tools (if available) to bring their sketches to life. This helps students connect math, art, and technology while practicing creativity, communication, and reflection.

Resources to print for this activity:

<https://www.freepik.com/vectors/2d-shapes-printable>

AI Doodle App:

<https://apps.apple.com/us/app/ai-doodles-from-sketch-to-art/id6744180745>

Chat GPT link: <https://chatgpt.com/>



Activity Instructions:

1. Introduce the Activity

Say:

“Today we’re going to be designers! You’ll use shapes to create something new that could help kids on the playground.”

- Show the shapes and ask students:

“What do these shapes remind you of?”

2. Distribute Materials

Give each student:

- A shape cut-out sheet
- Scissors
- Glue stick
- Open Glue here page in the book.
- Crayons or markers

3. Cut the Shapes

Remind students to cut carefully around each shape.

- Walk around to assist and encourage fine motor skills.

4. Explore and Imagine

Prompt students:

- “Look at your shapes before you glue—what can you make with them?”
- “Could one shape be a ramp, a flower, a sign, or a slide?”

Let students turn shapes different ways to spark new ideas.

5. Design and Glue

Have students arrange their shapes on the blank paper to make their creative structure or playground element.

- Once satisfied, they glue them down.
- Encourage them to give it a name or describe what it does.

6. Add Color and Details

Students use crayons or markers to:

- Add pathways, kids playing, labels, or scenery
- Use bright colors to make their ideas clear

7. Share and Reflect

Invite students to:

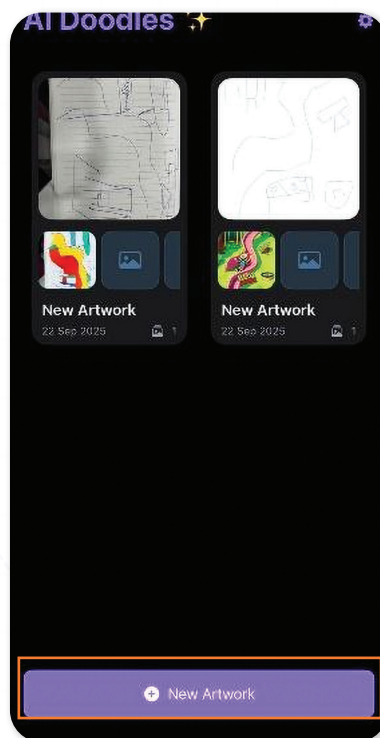
- Pair-share with a classmate to describe what they built
- Say how their idea helps all kids have fun or feel safe
- Possible shapes:

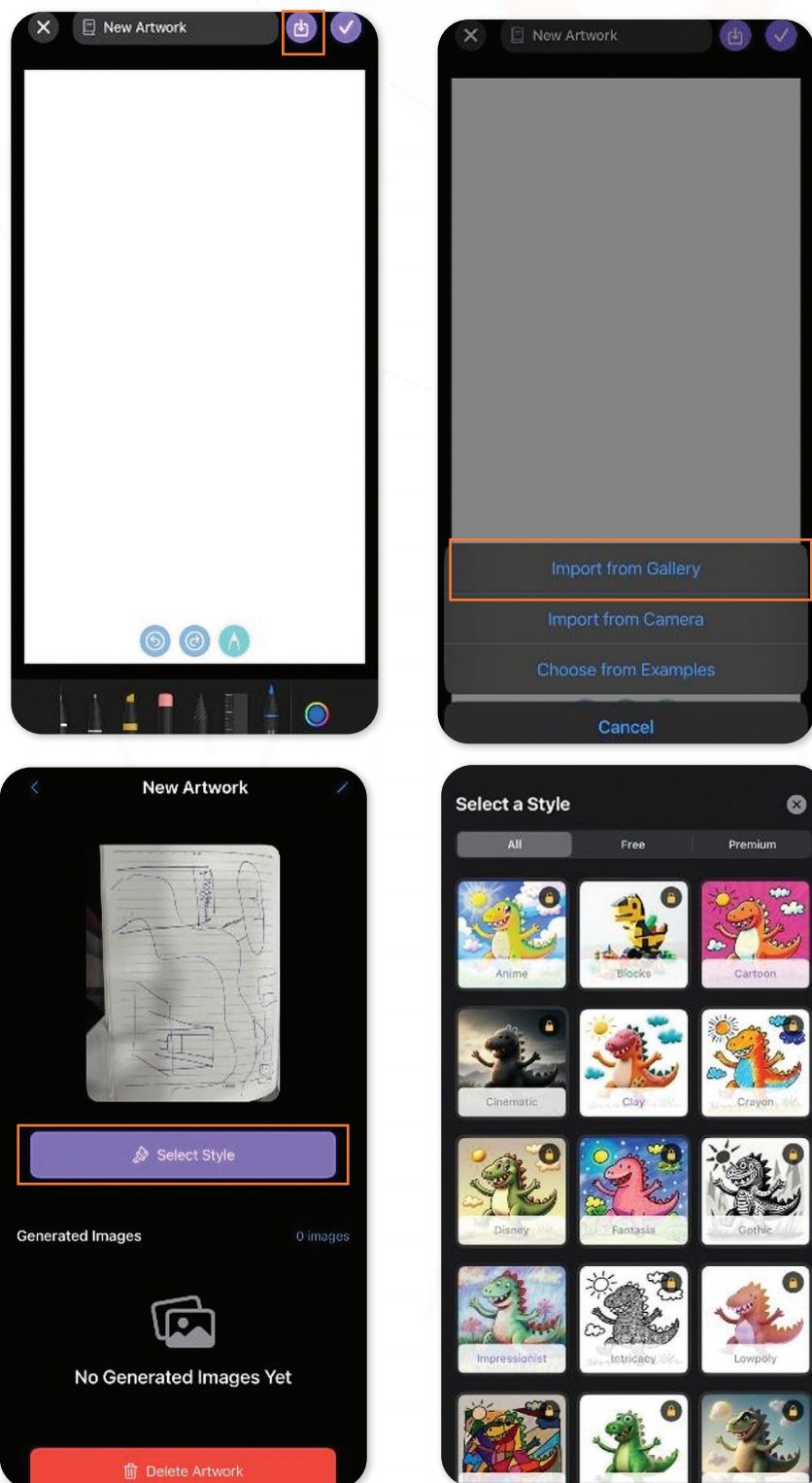


Extend with AI

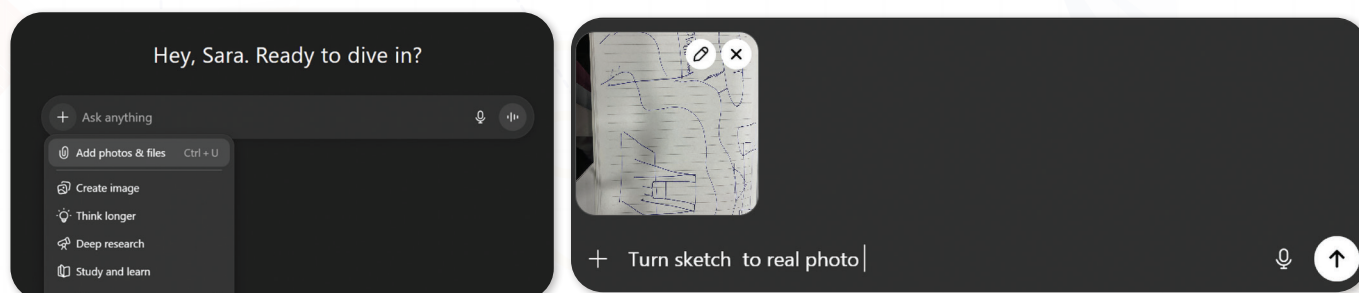
If time and technology are available, guide students to bring their sketches to life using AI tools:

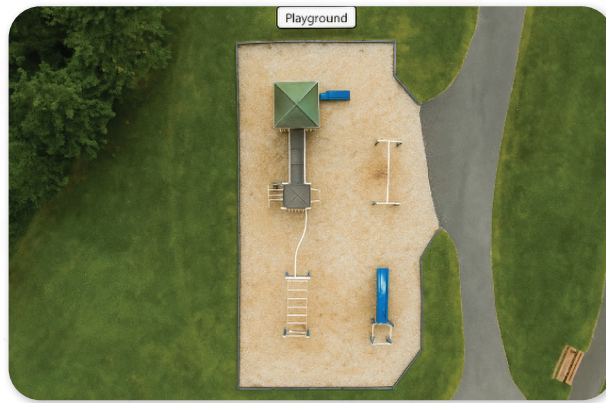
- With iPhone or iPad available: Use the AI Doodles app to upload or sketch their design and see how AI transforms it into digital art.





If devices are not available: Use ChatGPT with teacher support. Students describe their playground design, and the teacher (or group) generates a fun visualization together.





Possible Strategies for implementation:

1. Shape Brainstorm Chart

Create a simple anchor chart with examples:

- Triangle = ramp or roof
- Circle = flower or wheel
- Rectangle = bench or slide

2. Think-Pair-Build

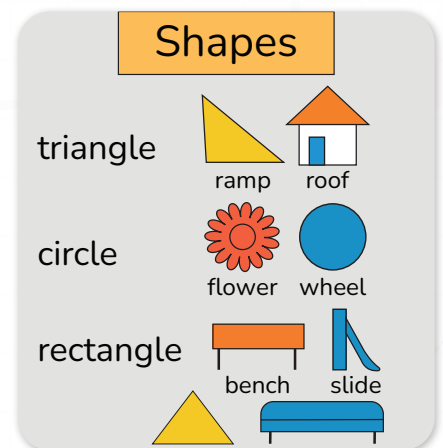
before gluing:

- Students talk with a partner:
“What are you going to make? What shapes are you using?”

3. Shape Reuse or Layering Challenge

Encourage students to combine shapes:

- “Can you use 2 shapes to make 1 new thing?”



Explorer's Toolkit



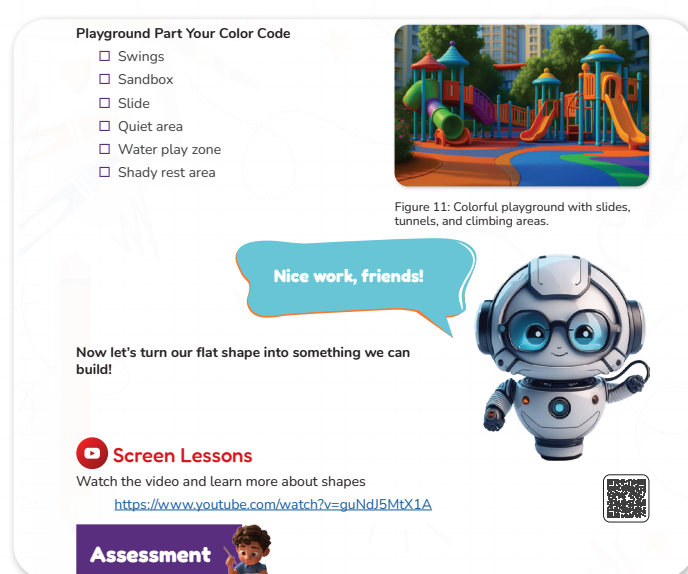
Teacher Role:

- **Introducer:** Introduce the concept of using colors as a code to help people navigate spaces, relating it to real-world examples like traffic lights or maps.
- **Facilitator:** Guide students in understanding the purpose of a playground color code – to help everyone, including those with visual impairments or cognitive differences, find their way and understand the function of different areas.
- **Encourager:** Encourage creative and thoughtful color choices, prompting students to consider why certain colors might be helpful for specific areas.
- **Clarifier:** Ensure students understand the different playground parts listed (swings, sandbox, slide, quiet area, water play zone, shady rest area).
- **Sharer:** Provide an opportunity for students to share their color codes and explain their reasoning.

Expected From Students:

- **Understanding the Concept:** Students will grasp the idea that colors can represent different things and help with navigation.
- **Color Association:** Students will think about which colors they associate with different playground activities or areas.
- **Creative Decision-Making:** Students will choose a unique color for each listed playground part.
- **Hands-on Participation:** Students will color the squares next to each playground part to create their personal color code.
- **Articulation of Reasoning:** Students will be able to explain why they chose specific colors for each area.

Parts of the books included.



DIY Lab

Objective: Students will explore how colors can be used as visual guides in designing inclusive playgrounds. By creating a personal color code for different playground areas (e.g., swings, sandbox, quiet area), students will practice categorization, visual mapping, and spatial awareness, while learning how design choices can help everyone feel safe, included, and independent on the playground.

Activity Instructions

Step 1: Set the Context

- Say:

“Now we’re going to be color designers! Colors can help kids know where to go on the playground. Just like signs, colors can help show where it’s safe to swing, rest, or play.”

- Show the image of the playground and point out how different areas serve different purposes.

Step 2: Explain the Task

- Read the directions on the worksheet with the class.
- Say:

“You’re going to create your own Playground Color Code by choosing one color for each part of the playground. You’ll color the boxes next to each area.”

Step 3: Distribute Materials

- Hand out:
 - Crayons or colored pencils

Step 4: Independent Work

- Students choose a unique color for:
 - Swings
 - Sandbox
 - Slide
 - Quiet area
 - Water play zone
 - Shady rest area
- They color the square next to each label.

Optional: Let them draw a small key or symbol next to the color if they want to make it extra clear (e.g., a wave for water play).

Possible Strategies for implementation:

1. Anchor with Real-World Examples

- Show real images of color-coded playgrounds, public parks, or maps.
- Ask:

“Have you seen colors on signs or paths before? What do they tell us?”

2. Color Meaning Brainstorm

- Create a chart with students about what different colors can mean:
 - Blue = water
 - Green = nature or rest
 - Red = active play
 - Yellow = sunshine or excitement

3. Think-Pair-Share before Coloring

- Have students turn to a partner and talk about:

“What color do you think is best for the quiet area?”

“What’s a calm color? What’s a loud color?”

4. Flexible Expression Options

- Allow students to:

- Use crayons, markers, or stickers
- Add a symbol next to the color box (e.g., sun, swing, flower)

5. Teacher Modeling

- Fill in one example on the board:

“I’m going to make the sandbox yellow because sand is yellow. What do you think?”

6. Movement-Based Matching Game (Optional Extension)

- Use color cards and place them around the room.
- Call out a playground area (e.g., “Quiet Zone”), and students move to the color they would choose for it.

7. Reflection and Sharing

- After coloring, students can:
 - Present their choices to the class or a partner
 - Explain how their color code would help others



Teacher Role:

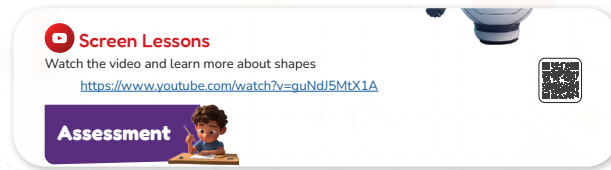
- **Introducer:** Introduce the concept of 2D and 3D shapes and how they relate to building.
- **Facilitator:** Guide students through the pre-, during-, and post-watching activities.
- **Purpose Setter:** Clearly explain what students should be looking for in the video regarding how flat shapes form 3D objects.
- **Engagement Encourager:** Prompt students to actively observe and think critically while watching.
- **Discussion Moderator:** Lead the post-watching discussion, ensuring all students have an opportunity to share their observations and insights.
- **Concept Clarifier:** Address any questions or misunderstandings that may arise after watching the video, particularly about 2D and 3D shapes.
- **Connection Maker:** Help students connect the information in the video to the playground design activities and the shapes they have been working with.

Expected From Students:

- **Active Viewing:** Students will watch the video attentively, focusing on how flat shapes are used to construct 3D objects.
- **Observation:** Students will identify the flat shapes shown in the video and the 3D objects they form.
- **Thoughtful Participation:** Students will contribute meaningfully to the pre- and post-watching discussions, sharing their initial thoughts, observations, and reflections.

- **Critical Analysis:** Students will think critically about the relationship between 2D and 3D shapes in the context of building.

Parts of the books included.



Screen Lessons

Objectives: Students will explore how 2D shapes (like circles, squares, and triangles) are the building blocks for creating 3D objects (like spheres, cubes, and pyramids). By watching the video, participating in a shape-spotting challenge, and discussing what they observed, students will build spatial awareness and begin to understand the role of flat shapes in real-world structures and design.

Video link: <https://www.youtube.com/watch?v=guNdJ5MtX1A>



Possible Strategies for implementation:

1. Pre-Watching: Set a Purpose

- Strategy: Brainstorm & Categorize (2D vs. 3D)
 - How to Implement:
 1. On the board, draw two columns: 2D Shapes and 3D Objects.

Ask:

“What flat shapes do you know?” (Add to the 2D column: circle, square, triangle, rectangle)

Then ask:

“What things can we build that are not flat?” (Add to the 3D column: cube, ball, cone, pyramid)

Tell students:

“We’re going to watch a video that shows how these flat shapes help us make real, 3D things!”

Benefit: Activates background knowledge, prepares students to make connections between flat and solid shapes.

2. During Watching: Active Engagement

- Strategy: Shape Spotting Challenge
 - How to Implement:
 1. Give each student a small “Shape Detective” sheet or let them tally on whiteboards.
 2. Ask them to watch closely and make a tally mark every time they see a flat shape being used (like a triangle or circle).

3. Also, challenge them to name or sketch the 3D shape it became (e.g., square → cube, circle → cone).

Benefit: Builds attention to detail, reinforces visual learning, and helps link 2D to 3D transformation.

3. Post-Watching: Discussion and Reflection

- Strategy: "I Saw... So It Can Build..." Sentence Frame
 - How to Implement:
 1. Ask students to complete and share the sentence:
"I saw a [2D shape], so it can build a [3D object]."
Example: "I saw a circle, so it can build a cylinder!"

Benefit: Reinforces understanding, encourages vocabulary use, and gives students confidence in applying new knowledge.

Elaborate



Teacher Role:

- **Clarifier:** Ensure students understand the difference between 2D (flat) shapes and 3D (solid) shapes, revisiting prior lessons or the video as needed.
- **Facilitator:** Guide students in using Tinkercad or other digital tools to explore and build with 3D shapes.
- **Encourager:** Support creative thinking by reminding students of the goal—designing a chair for families that is both fun and functional.
- **Supporter:** Help students navigate basic functions in Tinkercad (e.g., resizing, rotating, combining shapes).
- **Observer:** Watch how students identify shapes, choose shapes that roll or stack, and apply this knowledge while designing. Take note of those who show understanding and those who need further support.

Expected From Students:

- **Shape Identification:** Students will correctly identify examples of 2D and 3D shapes and describe their features (e.g., "a circle is flat," "a sphere can roll").
- **Shape Application:** Students will predict which 3D shapes can roll, stack, or support weight—connecting this knowledge to design thinking.
- **Digital Modeling:** Students will use Tinkercad to create a simple 3D model of a small chair using basic shapes like cylinders, cubes, and rectangles.
- **Creative Design Thinking:** Students will show their understanding of shape properties by applying them purposefully in a model meant for real-world use (seating).

Parts of the books included.

Draw Your Color Code!
Look at the boxes below. Choose your own color for each area and color the square.

Playground Part Your Color Code

- ☐ Swings
- ☐ Sandbox
- ☐ Slide
- ☐ Quiet area
- ☐ Water play zone
- ☐ Shady rest area

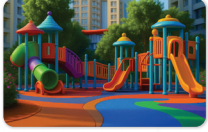


Figure 11: Colorful playground with slides, tunnels, and climbing areas.

Nice work, friends!

Now let's turn our flat shape into something we can build!

Screen Lessons
Watch the video and learn more about shapes
<https://www.youtube.com/watch?v=guNdJ5MX1A>

Assessment

Apprentice
Circle the correct answer:
Which one is a 2D shape?



Cube Sphere Circle Cone

STEAM SCOUTS 19

Builder
Make a small chair on Tinkercad for families and people to sit on in the playground.

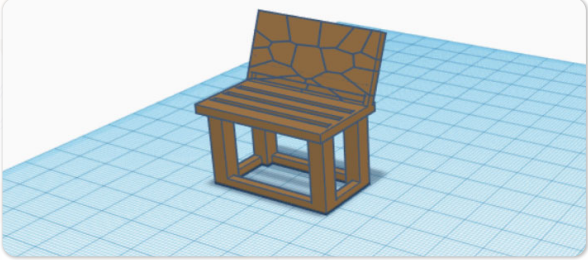


Figure 12: A playground resting chair designed in Tinkercad.



Apprentice

Objective: Students will demonstrate their understanding of the differences between 2D and 3D shapes by identifying which shape is flat (2D) and which shape can roll (3D). This activity reinforces foundational geometry concepts, supports shape recognition, and helps develop critical thinking related to the properties and functions of basic geometric forms.

Answer:

Apprentice
Circle the correct answer:
Which one is a 2D shape?



Cube Sphere Circle Cone

Which shape can roll?



Square Triangle Sphere Rectangle

Possible Strategies for implementation:

1. Shape Sort Warm-Up

- Start with a hands-on shape sort using physical cutouts or 3D models.

- Ask students to sort into “Flat (2D)” and “Solid (3D)” piles.

2. Think Aloud Modeling

- Model how to analyze one image aloud:
 “Let’s look at the circle. It’s flat—so that’s a 2D shape!”
 “The sphere is round all over and rolls—it must be 3D!”

3. Partner Discussion Before Circling

- Pair students and have them discuss each question together before circling:
 “Which do you think is 2D? Why?”
 “Can you show me how this shape would roll?”

4. Use Real Objects for Demonstration

- Bring in real items like a ball (sphere), a cube block, or a paper circle.
- Let students test which ones roll to reinforce their predictions from the worksheet.

5. Interactive Display on the Board

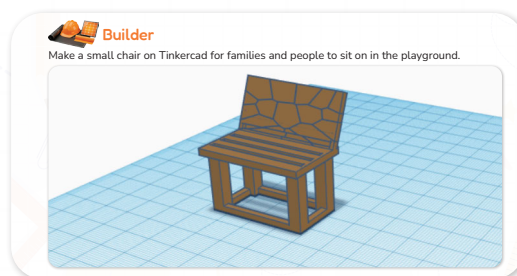
- Project the worksheet and complete one part together as a class using markers or digital tools.
- Invite volunteers to come up and circle answers.

6. Shape Movement Game (Extension)

- Ask students to pretend to “be” shapes:
 “If you are a sphere, roll across the rug!”
 “If you are a square, stand still and wide!”



Objective: Students will apply their understanding of 3D shapes and spatial reasoning to design a small digital chair using Tinkercad. Through this activity, students will explore how shapes can be combined and manipulated to create functional objects for real-world use—specifically, seating for families in an inclusive playground. The activity encourages design thinking, creativity, and digital modeling skills.



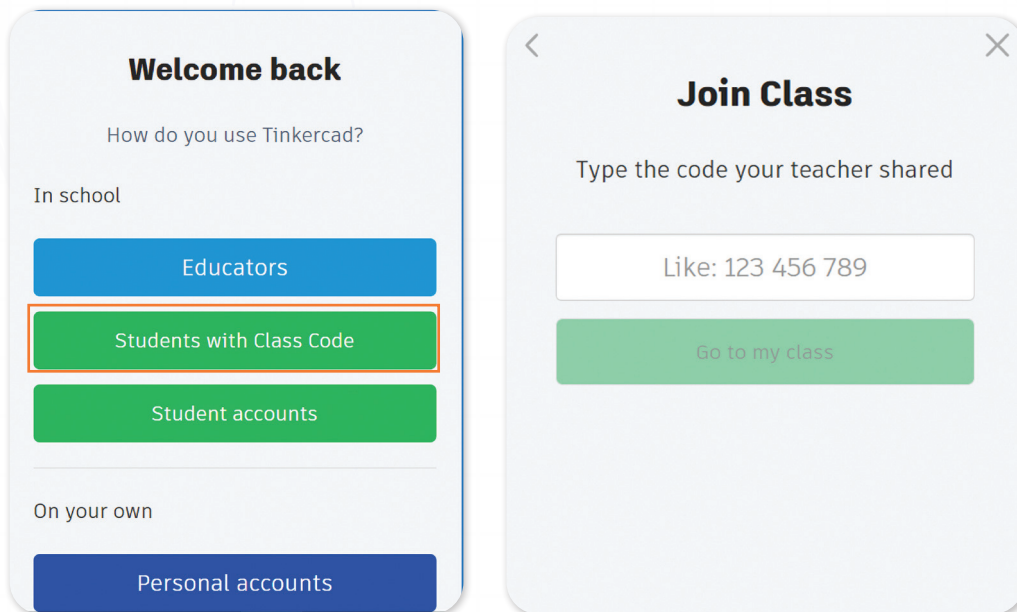
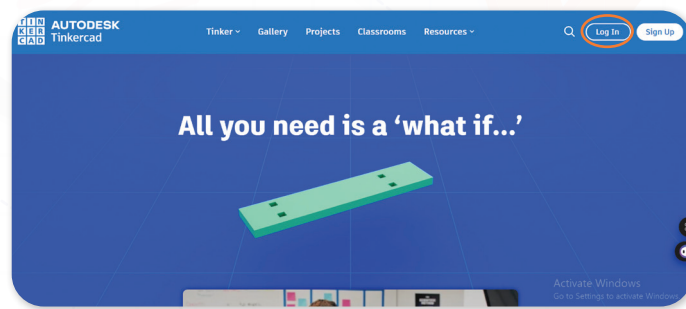
Software link:

<https://www.tinkercad.com/>

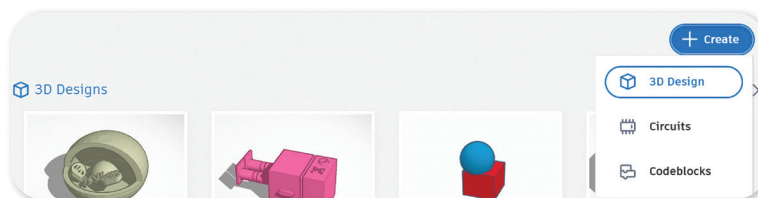
Activity Instructions



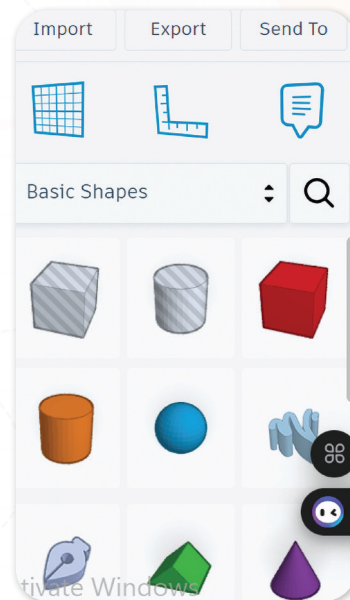
1. Access Tinkercad: Open the Tinkercad website or app and log in (students need class code and username should be pre-prepared).



2. Create New Design: Start a new 3D design.

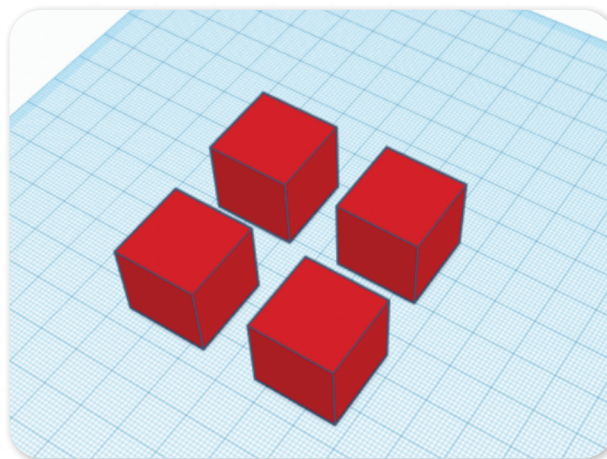


3. Basic Shapes: Introduce or remind students about the basic shapes available in Tinkercad (e.g., boxes, cylinders, spheres).

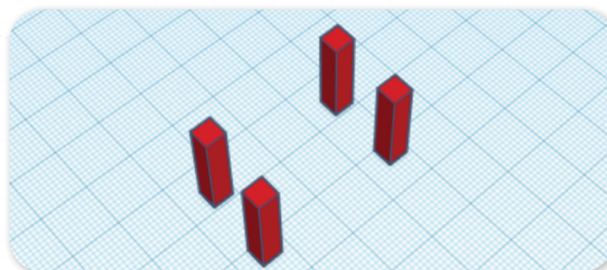


4. Build the Chair: Guide students to use these basic shapes to construct a simple chair. This might involve:

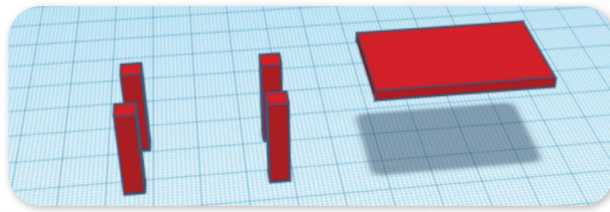
- The legs: Start with four blocks and place them so they are lined up with each other in a rectangle formation.



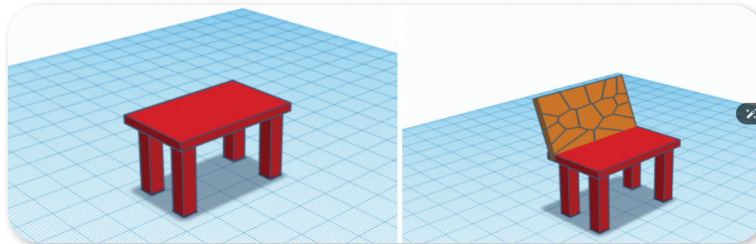
- Shaping the Legs: Shape down the legs so they are 4 wide, 4 long, and 20 high. Make the rest of the blocks these same dimensions. Then, place the two front legs 30 units apart and the back legs 30 units apart. The left front and back legs should be 15 units away, same with the right side.



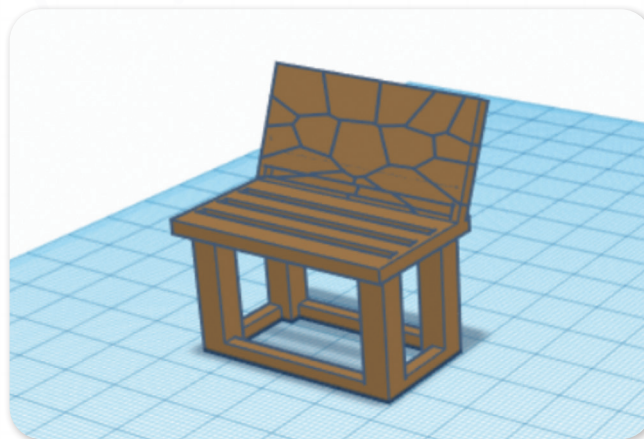
- Create the Sitting Platform: Create a block and mold it to these dimensions, 23 wide x 38 long x 2 high.



- Placing the Platform and Adding a Back: Place the platform so that it is in the middle, the side should hang over a little bit. Find a back piece that you like and mold it to the same dimensions as the sitting platform. Then, rotate it to a 67.5 degree angle and place it on the bench, it should line up with the platform.



- Paint, and Personalize Your Bench: Now you can group all the pieces together or paint the individually, i also added groovez like a real bench but you don't have to.



- Reflection Questions:
 - What shapes did you use to build your chair?
 - Why did you choose those shapes?
 - How is your 3D chair different from a drawing of a chair?



Mastermind

Objective: Students will explore real-world objects in their environment to identify whether they are 2D or 3D shapes and use a ruler to measure their height or length accurately. This activity supports the development of geometry and measurement skills, enhances students' ability to classify shapes, and reinforces the importance of precise measuring using standard tools.



Let's Try It!

- Look around the classroom for some shapes (a block, a book, or even a box).
- Use your ruler to measure how long or tall each one is.
- Write down your measurement.

Activity Table:

Object or shape	2D or 3D?	My Measurement
	☐ 2D ☐ 3D	
	☐ 2D ☐ 3D	
	☐ 2D ☐ 3D	

Don't forget: Always start measuring at 0!

Materials needed: Ruler

Activity Instructions

1. Introduce the Task:

"Now, we're going to be shape explorers and measurement masters! We're going to look around our classroom for different things and see if they are flat like a picture (2D) or solid like a block (3D). Then, we'll use our rulers to see how long or tall they are."

2. Explain 2D vs. 3D:

Briefly review the difference between 2D shapes (flat, like a circle drawn on paper) and 3D shapes (solid, like a ball or a box). You can use examples from the classroom.

3. Demonstrate Measurement:

Show students how to use a ruler correctly, emphasizing starting at the zero mark and reading the measurement along the edge. Demonstrate measuring the length of a book or the height of a block.

4. Guide Object Selection:

Encourage students to choose a variety of objects (e.g., a flat piece of paper, a book, a pencil case, a building block).

5. Distribute Activity Sheets and Rulers:

Provide each student with the "Let's Try It!" activity sheet and a ruler.

6. Circulate and Support:

Walk around the classroom as students work, helping them identify shapes, use the rulers correctly, and record their data. Offer assistance as needed.

7. Share and Discuss (Optional):

After students have completed their measurements, you can have a brief class discussion where they share what objects they measured, whether they were 2D or 3D, and their measurements.

Possible Strategies for implementation:

1. Model the Process First

- Use a familiar object (e.g., a book or ball) and demonstrate:
 - How to decide if it's 2D or 3D
 - How to line up the ruler and measure from 0
 - How to record the object and measurement in the table

2. Use a Shape Hunt Anchor Chart

- Create a visual with real classroom objects sorted into 2D and 3D categories (e.g., "poster = 2D", "block = 3D").
- Keep it posted for students to reference during the activity.

3. Assign Shape Hunt Buddies

- Pair students for collaboration: one holds the ruler, the other records.
- Encourage them to discuss their choices:
“Why do you think this is 3D?”
“How do we measure this side?”

4. Use Flexible Tools

- Offer both standard rulers and nonstandard units (e.g., paper clips or cubes) for differentiated learners.
- Allow drawing the object and writing "about this long" if exact numbers are too challenging.

5. Incorporate Movement

- Turn the activity into a “Shape and Measure Walk” around the room.
- Students move with clipboards and rulers to find and measure 3 items.

6. Reflection & Sharing Circle

- After the activity, bring students together to discuss:

“What was the longest object you measured?”

“Was anything trickier to measure than you thought?”

Evaluate



Teacher Role:

- **Facilitator:** Organize a "Playground Showcase" where students can present their Tinkercad playground designs.
- **Observer:** Observe students as they present, noting their design choices and explanations.
- **Questioner:** Ask clarifying questions to encourage students to articulate their design decisions and how they incorporated inclusivity.
- **Feedback Provider:** Offer constructive feedback based on the rubric, highlighting strengths and areas for potential improvement.
- **Celebrator:** Acknowledge and celebrate the creativity and effort of each student.

Expected From Students:

- **Project Completion:** Students will have created a digital model of their inclusive playground in Tinkercad, including various play areas and the chair they designed.
- **Presentation:** Students will present their playground design to the class (or in small groups), explaining the different features and why they included them.
- **Articulation of Design Choices:** Students will be able to explain how their design promotes inclusivity and addresses the needs of all children.

- **Reflection:** Students will reflect on their design process and what they learned about creating accessible play spaces.

Parts of the Books Included:

Showcase

Use Tinkercad to design the space for your playground. Plan out what you want to include in each area, and remember to place the chair you created!



Figure 14: A playground design created in Tinkercad with slides, swings, trees, and a resting chair.



Showcase

Objective: Students will apply their understanding of inclusive playground design, spatial planning, and 3D modeling by creating a digital or physical layout of a playground. This includes organizing features such as play zones, quiet spaces, and their custom-designed Tinkercad chair. The activity reinforces design thinking, creativity, and communication as students explain the purpose of each area in their final design.

Software link: <https://www.tinkercad.com/>

Activity link: <https://url-shortener.me/4EIN>

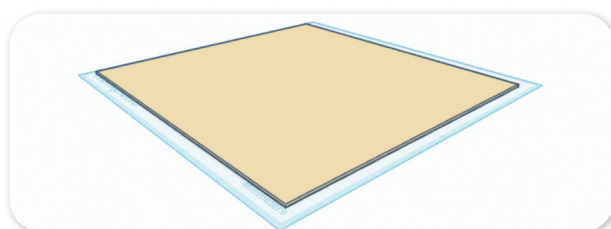
Activity Instructions

- Set Up Your Space:

The same steps you made in the builder activity to login the program.

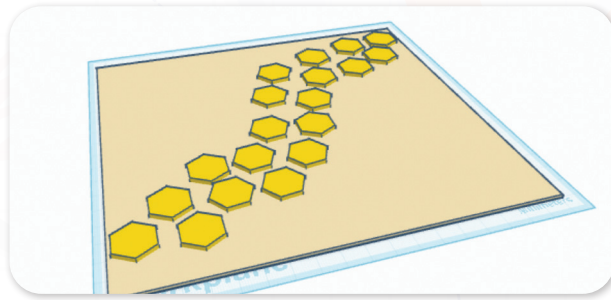
- Plan Your Layout:

1. Start to add the space where you will build your playground on, drag a box and change how wide and tall it is to fill the space so you can add other things to it and also change the color.

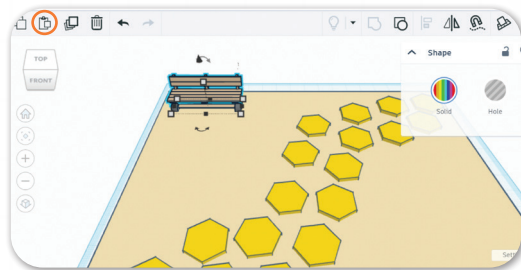
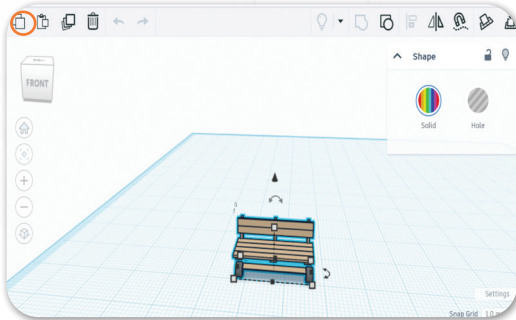


2. Start to think where you are going to add the spiral ramp roadn, drag hexagonal shape add it, duplicate it many times.

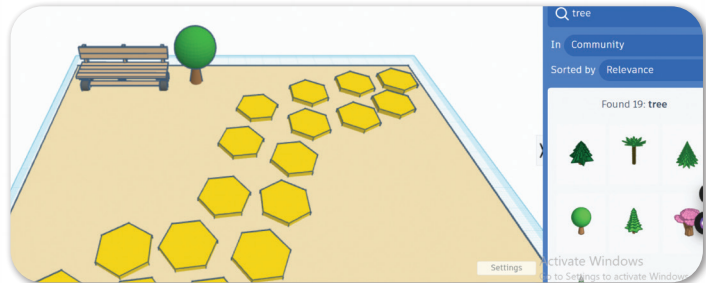
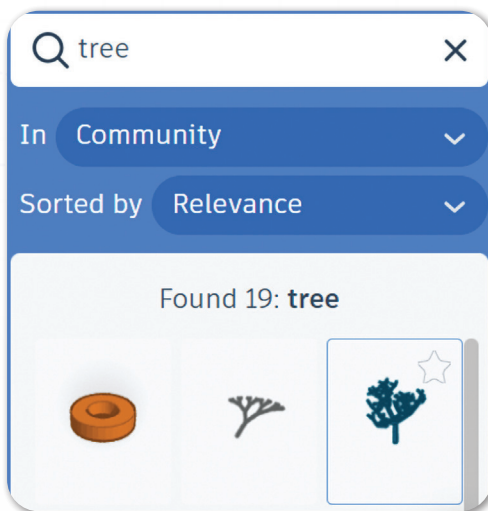




3. Add Your Chair Design: Copy the design that you already made before and past it to your design, then resize it.

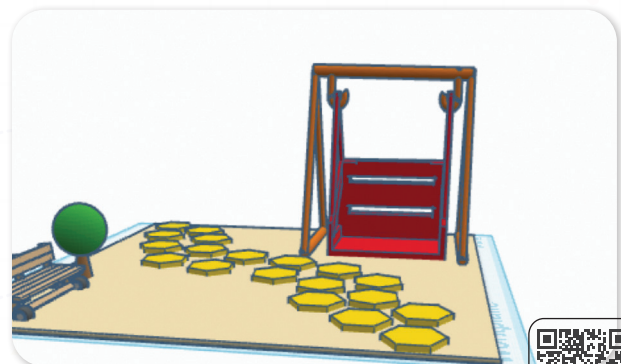
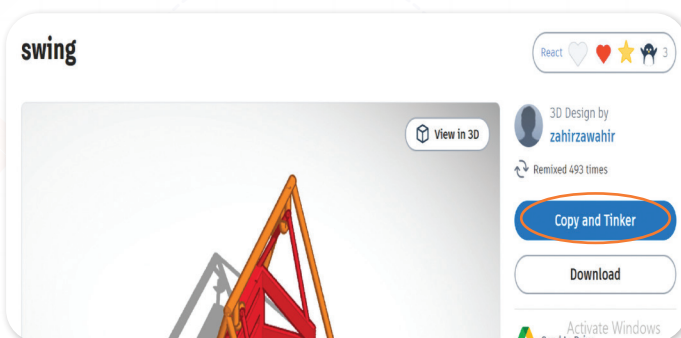


4. Use the search tool to search for a tree and add it to your space.



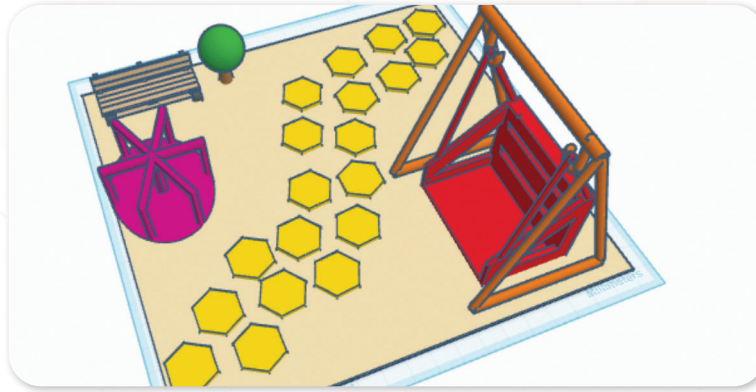
5. Help the students add a wide swing using tinker this, search on google for swing, click tinker this, copy it and past it to your space.

Link for the swing: <https://www.tinkercad.com/things/5DUXOQuWs3m-swing>



6. Repeat the same step and add the marry-go round.

Link for the marry-go round: <https://www.tinkercad.com/things/2W5hVJ1pHEy->



- Present or Label Your Design:
Add labels or be ready to explain:
“Why did I put this here?”
“Who is this part for?”
“How does this help everyone play?”

Possible Strategies for implementation:

1. Use a Design Template or Grid (Digital or Paper)

- Provide students with a base template or playground map grid.
- They can build directly in Tinkercad or draw/color in a top-down layout.

2. Revisit Prior Design Decisions

- Remind students to include features from earlier activities:
 - The chair from Tinkercad
 - Their shape-based playground parts
 - Color-coded areas (quiet zone, sandbox, etc.)

3. Use Peer Review or Partner Check-In

- Before finalizing, students can share their layout with a buddy and ask:
 - “Can you find the quiet space?”
 - “Is there a place for everyone?”

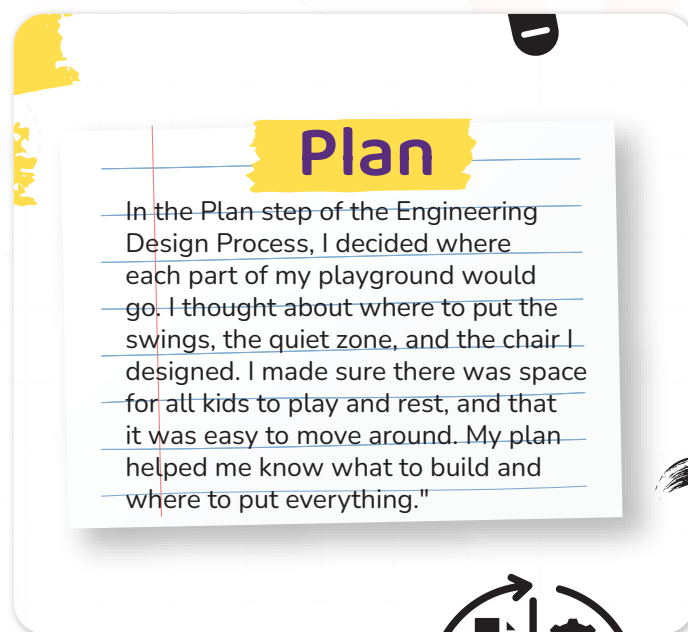
4. Guided Reflection Prompt

- After designing, ask:
 - “What part of your playground makes it fun for all kids?”
 - “Where can someone rest, play, or feel calm?”

Showcase: Rubric for Evaluating Playground Designs

Criteria	Beginning (1 Point)	Developing (2 Points)	Proficient (3 Points)
Inclusion (Accessibility)	Design shows limited consideration for diverse needs. Few or no accessible features are evident.	Design shows some consideration for diverse needs. At least one accessible feature is included.	Design clearly considers diverse needs. Multiple accessible features are thoughtfully integrated (e.g., ramps, varied surfaces, quiet area).
Use of Shapes	Limited or unclear use of different shapes. Shapes are used randomly or without purpose.	Shows some variety in shape use. Attempts to represent playground elements with shapes are evident.	Effectively uses a variety of shapes to clearly represent different playground elements (e.g., rectangular slide, circular swing area).
Incorporation of Chair	Chair is missing or not clearly integrated into the playground design.	Chair is present but may not be logically placed within the playground.	Chair is thoughtfully placed as a functional element within the playground design.
Presentation & Explanation	Student struggles to explain their design or the purpose of different elements.	Student provides a basic explanation of their design but may not clearly articulate inclusivity aspects.	Student clearly explains their design choices and articulates how their playground aims to be fun and accessible for everyone.
Effort & Creativity	Minimal effort is evident. Design lacks originality.	Shows some effort and a few original ideas.	Demonstrates significant effort and creative thinking in the overall design.

Don't forget to go back to the EDP page and add these to planning part.



Now I Can

Objective: Students will reflect on their learning by identifying the skills they used throughout their playground design project, including their understanding and use of 2D/3D shapes, color and size selection, digital design tools (Tinkercad), and measurement in centimeters and meters. This activity promotes metacognition, helps students take ownership of their learning, and reinforces math, design, and technology integration.

Activity Instructions

- **Review Each Statement:**

Read through the checklist one sentence at a time. You may ask your teacher if you're not sure what something means.

- **Check What You Can Do:**

Put a check mark in the box next to the things you feel confident about or remember doing.

- **Think Back to Your Work:**

Look at your playground model, shape designs, or Tinkercad project to help you remember what you've done.

- **Optional Reflection Prompt (Teacher-Directed):**

After completing the checklist, answer one of these in your notebook or aloud:

- "The thing I'm most proud of is..."
- "One thing I want to get even better at is..."
- "I used..... shape when I built my....."

- **Class Discussion (Optional):**

Share one thing you learned or enjoyed from the project with the class or a partner.

Relation to Standards:

1. Next Generation Science Standards (NGSS):

- **K-2-ETS1-2:** Develop a simple sketch, drawing, or physical model to illustrate how the shape of an object helps it function as needed to solve a given problem. (Students are designing a functional space using shapes.)
- **K-2-ETS1-3:** Analyze data from tests of two objects designed to solve the same problem to compare the strengths and weaknesses of how each performs. (Implicitly, students are designing solutions to the problem of inaccessible playgrounds.)

2. Common Core State Standards (CCSS) - English Language Arts (Speaking and Listening):

- **SL.1.4:** Describe people, places, things, and events with relevant details, expressing ideas and feelings clearly. (Students present and explain their playground designs.)
- **SL.1.5:** Add drawings or other visual displays to descriptions when appropriate to clarify ideas, thoughts, and feelings. (The Tinkercad model serves as a visual display.)

3. Common Core State Standards (CCSS) - Mathematics (Geometry):

- **1.G.A.1:** Distinguish between defining attributes (e.g., number of sides, number of vertices) versus non-defining attributes (e.g., color, size, orientation); build and draw shapes to possess defining attributes. (Students are using their understanding of shapes to construct their designs.)